THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. 4 g of hydrogen combines with 32 g of oxygen to form water. The mass of water formed and the ratio of hydrogen to oxygen are: 1

(a) 36 g, 1 : 4 **(b)** 28 g, 1 : 8 **(c)** 36 g, 1 : 8 **(d)** 36 g, 8 : 1

2. At 40 degrees Celsius, 100 g of water dissolves 64 g of potassium nitrate in a saturated solution. The mass percentage of the saturated solution is about: 1

(a) 36 per cent **(b)** 39 per cent **(c)** 64 per cent **(d)** 100 per cent

3. The angle between a ray and a plane mirror is 35 degrees. If the mirror is rotated by 15 degrees so that the angle of incidence increases, the new angle between the incident and the reflected ray is: 1

(a) 100 degrees **(b)** 70 degrees **(c)** 110 degrees **(d)** 140 degrees

4. A pole of height 3 m has a shadow of 4 m and a nearby tree has a shadow of 20 m at the same time. The height of the tree is: 1

(a) 20 m **(b)** 26 m **(c)** 15 m **(d)** 10 m

5. In the food chain grass, grasshopper, frog, snake, the energy available is least at the level of the: 1

(a) grass **(b)** snake **(c)** grasshopper **(d)** frog

6. The core of the Earth has a radius about 3400 km and the Earth about 6400 km. The ratio of the core radius to the Earth radius is about: 1

(a) 0.53 **(b)** 0.34 **(c)** 1.88 **(d)** 0.64

7. A thermometer reads 3 degrees C in melting ice. A reading of 53 degrees C for warm water corresponds to a true value of: 1

(a) 56 degrees C **(b)** 50 degrees C **(c)** 53 degrees C **(d)** 47 degrees C

8. A cell loses its nucleus. The most probable result is that the cell: 1

(a) stops functioning properly **(b)** grows faster

(c) becomes a plant cell **(d)** forms chloroplasts

9. A student claims, 'I took antibiotics for my cold and it got better, so antibiotics cure colds.' The flaw is: 1

(a) antibiotics are vaccines
(b) antibiotics cure all diseases
(c) colds are bacterial
(d) colds are viral and often recover by themselves

10. A coil of wire is wound on a plastic tube and then on an iron rod, with the same current. The magnetic effect is stronger with: 1

(a) the plastic tube **(b)** equal
(c) the iron rod **(d)** neither, as there is no magnetic effect

11. A tug-of-war rope is stationary. If team A suddenly pulls harder, then the rope: 1

(a) breaks always **(b)** moves towards team B
(c) stays at rest **(d)** moves towards team A

12. Two cyclones, A over warm sea and B moving over land, are compared. Which statement is right? 1

(a) A is likely to be stronger **(b)** B is likely to be stronger
(c) both are equal **(d)** neither has wind

13. A student observes that a drop of food colour spreads equally in a glass of still water but faster in warm water. The explanation is: 1

(a) gravity acts differently
(b) water is thicker
(c) temperature increases the motion of particles
(d) colour is lighter

14. A coin is made of an alloy of copper and nickel. This coin is a: 1

(a) compound **(b)** element
(c) heterogeneous mixture **(d)** homogeneous mixture

15. A liquid X mixes with water in all proportions while liquid Y does not. Which can be separated from water with a separating funnel? 1

(a) X **(b)** Y **(c)** both **(d)** neither

16. Light incident on a smooth metal surface at 90 degrees to the surface is reflected at: 1

(a) 45 degrees
(b) 180 degrees
(c) 0 degrees
(d) 90 degrees to the surface (back along normal)

17. The Moon appears the same size as the Sun in the sky although the Sun is about 400 times larger because the Sun is also: 1

(a) smaller **(b)** dimmer
(c) about 400 times farther **(d)** closer

18. Which of the following is a good example of mutualism? 1

(a) mosquito and human **(b)** tapeworm and human
(c) tiger and deer **(d)** bees and flowers

19. A glacier's melting adds water to the oceans. If all land ice melted, the effect on coastal cities would be: 1

(a) flooding by rising sea levels **(b)** cooler climate

(c) no change **(d)** drought on the coast

20. Assertion (A): An experiment should be repeated several times. Reason (R): Repetition ensures that the hypothesis is always true. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Explain why a gas does not collect at the bottom of a container although it has weight. 2

22. 4 g of hydrogen reacts completely with oxygen forming 36 g of water. Find the mass of oxygen and the mass ratio of hydrogen to oxygen. 2

23. 100 g of a saturated solution at 60 degrees Celsius contains 37.5 g of solute. Find the mass percentage and the mass of water. 2

24. A ray incident on a mirror at 30 degrees to the surface is reflected. If the mirror turns through 10 degrees, find the possible new angle of incidence. 2

25. Why does a calendar need a leap-day rule with exceptions for century years? Give an example with 1900 and 2000. 2

26. Name the trophic levels in the food chain grass, deer, tiger and state the level that has the least energy. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Compare the solubility of NaCl (36 g/100 g water at 20 degrees Celsius) and KNO3 (32 g/100 g water at 20 degrees Celsius). If 100 g water at 20 degrees Celsius is given 40 g of each, find the undissolved mass for each. 3

28. Two plane mirrors are inclined at an angle theta. If 7 images are formed, find theta. 3

29. At a place the Sun is directly overhead at noon on 21 March. A pole of height 5 m is placed there. Describe the shadow at noon and at 3 pm. 3

30. Why are there fewer top carnivores than herbivores in an ecosystem? Give three points. 3

31. Describe how Earth's layers formed by differentiation and what evidence supports a liquid outer core. 3

OR

From a graph, how do you judge whether two quantities are directly proportional? 3

32. Name the processes carried out by Amoeba that are done by organs in a human. 3

OR

Write the diseases caused by 3

(a) iodine

(b) vitamin D

(c) iron deficiency and the food source to prevent each.

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A concave mirror has a radius of curvature of 20 cm. Find its focal length. Where would a parallel beam from the Sun meet after reflection? Why is such a mirror used in a torch? 5

34. Describe a long-term investigation to record the Sun's noon shadow length and direction through the year and explain the expected results with the solstices and equinoxes. 5

35. The population of lions in a forest dies due to disease. Trace the consequences through the food web with at least five steps in the ecosystem and the final outcome. 5

36. The Earth's radius is 6400 km. Using the crust 35 km, mantle 2900 km, outer core 2200 km, find the radius of the inner core, the percentage of radius for each layer and comment on which layer is thickest. 5

OR

Discuss the role of curiosity, observation and persistence with reference to the work of scientists who repeated experiments many times. 5

37. Explain why some cells are specialised. Give two examples of how shape suits function. 5

OR

Explain the concepts: pathogen, host, vector, symptoms, immunity, vaccine, antibiotic, epidemic using one line each. 5

38. Compare how a cell, a battery and mains supply differ in voltage and risk. Explain why a cell is safe to touch but the mains is not. 5

OR

Discuss how the shape of a body affects air and water resistance and give five examples of streamlining in nature and technology. 5

END OF QUESTION PAPER **Total: 80 Marks**

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